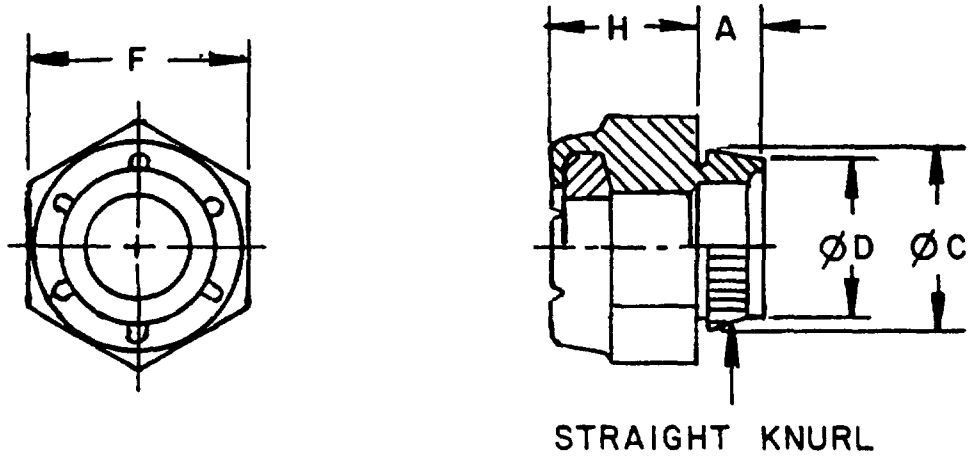
MILITARY SPECIFICATION SHEET

NUT, SELF-LOCKING, CLINCH (SWAGE-CLINCHING, KNURLED SHANK, 250° F, UNJF-3B)

This specification is approved for use by Naval Sea Systems Command (Ordnance Systems), Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the nuts described herein shall consist of this specification and the latest issue of MIL-N-45938.



NOTES:

1. All dimensions are in inches.

FIGURE 1. NUT, SELF-LOCKING CLINCH (SWAGE-CLINCHING, KNURLED SHANK, 250° F, UNJF-3B)

FSC 5310

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TABLE I. DASH NUMBERS AND DIMENSIONS.

DASH NUMBER			THREAD SIZE	A SHANK LENGTH ±.006	SHEET THICKNESS	HOLE SIZE IN SHEET		ØC ±.002	ØD +.000 -.003	F ±.010	H ±.015
CRES	AL ALLOY	STEEL				MAX	MIN				
1	1A	1G	.164-36 UNJF-3B	.085	.032 TO .053	.271	.268	.275	.267	.375	.250
20	20A	20G	.190-32 UNJF-3B	.085	.032 TO .053	.271	.268	.275	.267	.375	.250
2	2A	2G		.105	.054 TO .073						
21	21A	21G		.135	.074 TO .103						
22	22A	22G		.165	.104 TO .133						
23	23A	23G		.195	.134 TO .163						
3	3A	3G	.250-28 UNJF-3B	.135	.074 TO .103	.355	.352	.359	.351	.438	.313
4	4A	4G		.165	.104 TO .133						
5	5A	5G		.195	.134 TO .163						
6	6A	6G	.3125-24 UNJF-3B	.135	.074 TO .103	.412	.408	.416	.407	.500	.344
7	7A	7G		.165	.104 TO .133						
8	8A	8G		.195	.134 TO .163						

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REQUIREMENTS:

1. Material. Corrosion resistant steel, Type 303 (UNS S30300) in accordance with ASTM A581 or ASTM A582.
Aluminum alloy, Grade 2017-T4 (UNS A92017) or Grade 2014-T6 (UNS A92014) in accordance with ASTM B211.
Carbon steel, Grade 12L14 (UNS G12144) in accordance with ASTM A107, or Grade 4130 (UNS G41300) in accordance with ASTM A322, or Grade 10L18 (UNS G10180) in accordance with ASTM A510.
The Locking Insert shall be nylon.

2. Protective coating and surface treatment. Aluminum alloy shall be anodized in accordance with MIL-A-8625. Carbon steel shall be cadmium plated. Corrosion resistant steel shall be cleaned, descaled, and passivated.

3. Threads. Threads shall be in accordance with MIL-S-8879.

4. Push-out and torque-out. Push-out and torque-out values shall be as specified in Table II, when properly installed in aluminum alloy or low carbon steel sheet.

5. Part number. The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table I.

Example: M45938/9-4A

- 4A: Dash number
- M45938/9: Specification sheet number

M45938/9-4A indicates - Nut, Self-Locking, Clinch, (Swage-Clinching, Knurled Shank, 250° F). Aluminum alloy, .250-28 UNJF-3B, .165 Shank Length.

TABLE II. PERFORMANCE AND INSTALLATION DATA

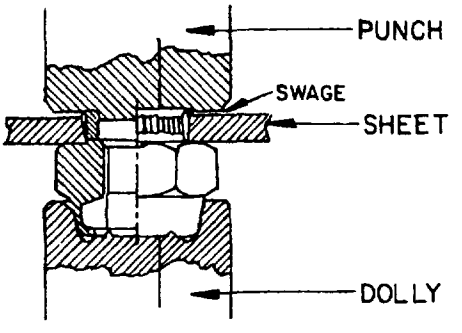
THREAD SIZE	PUSH-OUT LB - MIN	TORQUE-OUT IN - LB MIN	MAXIMUM INSTALLATION FORCE LB		
			AL NUT	CRES NUT	STEEL NUT
.164-36 UNJF-3B	140	20	2100	4000	2100
.190-32 UNJF-3B	200	30	2100	4000	2100
.250-28 UNJF-3B	250	50	4000	4300	4000
.3125-24 UNJF-3B	400	110	4500	4500	4500

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RECOMMENDED INSTALLATION PROCEDURE

a. Drill or punch hole in sheet (See Table I for size).
b. Using the installation set-up shown in the sketch below, insert nut shank into hole. Select the correct Dolly from Table III below. Place the Dolly on the nut crown and, using a light hammer blow or small press, force the nut into the structure. The nut base must be fully and squarely seated against the structure.
c. Using the Dolly to support the nut head, flare the edges of the shank outward using the correct Punch from Table IV below. Do not exceed the installation forces specified in Table II.
d. Install the screw or bolt from the shank end of the nut.

INSTALLATION SET-UP



DOLLY

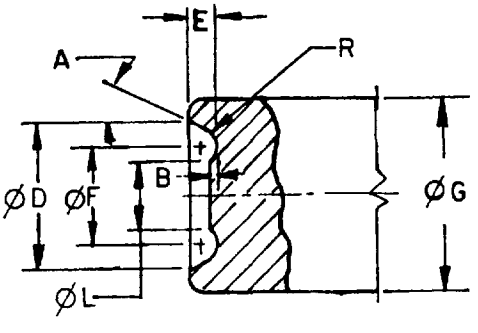


TABLE III. DOLLY DIMENSIONS

THREAD SIZE	A ±2°	B ±.002	ØD +.002 -.000	E ±.002	ØF REF	ØG REF	ØL ±.002	R ±.001
.164-36 UNJF-3B	23°20'	.011	.380	.070	.247	.500	.179	.055
.190-32 UNJF-3B	15°	.008	.376	.070	.267	.500	.213	.048
.250-28 UNJF-3B	25°	.010	.451	.080	.326	.563	.273	.040
.3125-24 UNJF-3B	20°	.012	.515	.075	.398	.625	.337	.045

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PUNCH

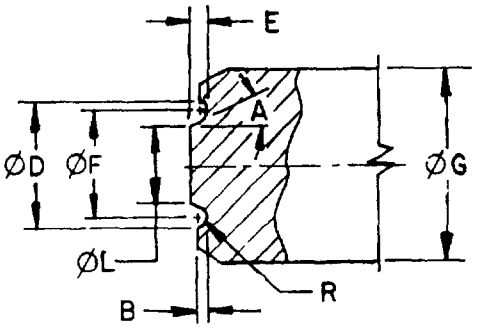


TABLE IV. PUNCH DIMENSIONS

THREAD SIZE	A ±2°	B ±.002	ØD ±.002	E ±.002	ØF REF	ØG REF	ØL ±.002	R ±.001
.164-36 UNJF-3B	25°	.014	.318	.038	.266	.500	.200	.023
.190-32 UNJF-3B	25°	.014	.318	.038	.266	.500	.200	.023
.250-28 UNJF-3B	30°	.012	.403	.043	.350	.563	.260	.035
.3125-24 UNJF-3B	25°	.015	.460	.050	.406	.625	.319	.032

Custodians:
Navy - OS
Air Force - 99

Review activities:
Army - ME
Air Force - 11, 82
NSA - NS
DLA - IS

User activities:
Navy - AS, MC, SH, YD

Agent:
DLA - IS

Preparing activity:
Navy - OS

(Project 5310-N040)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-N-45938/9A	NUT, SELF-LOCKING, CLINCH (SWAGE-CLINCHING) KNURLED SHANK, 250°F, UNJF-3B Project No. 5310-N040

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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